

Part 1: Controller Area Network (CAN) Fundamentals

The Foundation of Automotive Embedded Communication

1. The Historical Automotive Problem

Prior to the 1980s, vehicle electrical components used point-to-point physical wiring. Every single switch was wired directly to its corresponding actuator. As safety and comfort features (ABS, cruise control, electronic ignition, central locking) proliferated, vehicle wire harnesses exceeded several kilometers in length and 100+ kg in weight, introducing severe reliability, maintenance, and space issues.

In 1986, Robert Bosch GmbH invented the **Controller Area Network (CAN)** to replace point-to-point wiring with a shared, high-reliability, two-wire digital bus.

2. Physical Layer: Differential Signaling

A CAN bus operates using two balanced, twisted copper wires: **CAN High (CAN_H)** and **CAN Low (CAN_L)**, terminated at both ends with 120-ohm resistors (giving an overall nominal bus impedance of 60 ohms).

- **Recessive State (Logical '1'):** Both CAN_H and CAN_L sit at approximately 2.5V (Differential voltage = 0V).
- **Dominant State (Logical '0'):** CAN_H is driven to ~3.5V and CAN_L is pulled to ~1.5V (Differential voltage = ~2.0V).

Because the signal is differential, any electromagnetic interference (EMI) induces equal noise voltages on both lines, which completely cancels out at the receiver's differential amplifier. This provides exceptional noise immunity in harsh automotive environments.

***Key Takeaway:** A dominant bit ('0') always overrides a recessive bit ('1') on the physical bus. This electrical property is the bedrock of CAN's non-destructive bus arbitration.*

3. Non-Destructive Bitwise Arbitration

Unlike Ethernet (CSMA/CD), where simultaneous transmissions result in collisions and lost packets requiring random backoff delays, CAN uses **lossless bitwise arbitration**:

1. Every node listens to the bus while transmitting bit-by-bit.
2. If Node A transmits a recessive '1' but senses a dominant '0' on the wire, it immediately knows a higher-priority node is speaking.
3. Node A backs off instantly and transitions to listening mode without disrupting the winning frame.
4. The highest-priority frame (the one with the lowest numerical CAN ID) continues uninterrupted.

4. Standard 11-Bit Frame Structure (CAN 2.0A)

Field	Bits	Purpose
SOF (Start of Frame)	1	Dominant bit for node clock synchronization.
Identifier (ID)	11	Message priority and broadcast content identifier.
RTR	1	Remote Transmission Request (0=Data, 1=Remote Request).
Control (IDE, r0, DLC)	6	Data Length Code (0 to 8 bytes payload length).
Data Field	0-64	0 to 8 bytes of application payload data.
CRC & Delimiter	16	Cyclic Redundancy Check error detection.
ACK & Delimiter	2	Receiving nodes assert dominant bit to confirm receipt.
EOF (End of Frame)	7	7 consecutive recessive bits marking frame boundary.